Test

15 August 2025 20:43

Subject: Computer Applications **Max. Marks: 50**
Class: IX **Time: 1.5 hours**

Topics:

1. **Datatypes**
2. **Operators**
3. **Selection**
4. **Loops**

Section A contains 10 questions 2 marks each. You have to attempt all the questions

Section B contains 4 questions 10 marks each, you have to attempt only 3 questions

Section A

[Attempt all questions] *[2x10 =20]*

Q1. Convert the following into a while loop:

```
for (int i=10;i>4;i=i/2)
i=i-1;
System.out.println("Done");
```

Q2. Give and explain the output:

```
char ch='l';
int n=ch;
System.out.print(n+4);
```

Q3. Write a java statement to create an object named "Cricket" of a class "Sports".

Q4. What is an Infinite loop? Give an example of an infinite loop using while loop.

Q5. Write the output of the following code:

```
for (int i=5;i<20;i=i+3)
{
        if (i%2==0)
        System.out.println("J");
        if (i%4==0)
        System.out.println("B");
}
```

Q6. Write Java expression for the following:

$$\sqrt[3]{x^2} - \frac{y}{4z}$$

Q7. Write a statement to print the following including all the quotes and symbols:

He said, " Here's my book/pen ".

Q8. If x=4, y=2 then evaluate the following and find the final value of y:

```
y += ++x – x++ + --x
```

Q9. Write the corrected code and underline the corrections:

```
int x=4,y=6;
double z=math.power(x*2,y/3);
```

Q10.Arrange the following operators as per their precedence in ascending order:

( ) , != , % ,&&

Section B

[Attempt any 3 questions] **[3 x 10 = 30]**

The answers in this Section should consist of the Programs in Blue J environment or any program environment with Java as the base.

Q11. A **Tech Number** has even number of digits and if the number is split into two halves, then the square of sum of these halves is equal to the number itself. Write a program in Java to generate and print all four digit tech numbers.

Example:

Square of sum of the halves of 3025 = $(30 + 25)^2$
$=(55)^2$
$=3025$

Q12. Write a menu driven JAVA program to find the sum of the following series till mentioned term:

Choice 1

$S = \frac{x^3}{15} - \frac{x^6}{11} + \frac{x^9}{7} - \frac{x^{12}}{3}$ till 10 terms

Choice 2

$S = 2 + 5 + 10 + 17 + 26 + 37$ .....till T terms (T will be given by user)

Q13. Write a JAVA program to accept annual salary of an employee. The program will calculate and display the income tax paid by the employee based on the following chart:

Annual Salary	**Rate of Income Tax**
Up to 2,50,000	No Tax
2,50,001 to 5,00,000	10% of the amount exceeding 2,50,000
5,00,001 to 10,00,000	Rs 5000 + 20% of the amount exceeding 5,00,000
Above 10,00,000	Rs 10000 + 30% of the amount exceeding 10,00,000

Q14. Pronic number is a number which is the product of two consecutive integers, that is, a number n is a product of x and (x+1). Write a program in Java to check and print whether the number is **Pronic Number** or not.

Example:

Input :56

Output :Pronic Number

Explanation: 56 = 7 * 8 i.e 56 is a product of two consecutive integers 7 and 8.

****

```
package TestDoubts;
import java.util.*;


/**
 * Write a description of class Q14 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q14
{
```

N = 56

```
*/
public class Q14
{
    public static void main(String args[]){
        Scanner sc = new Scanner(System.in);
        int N, x, p = 1;
        System.out.print("\fEnter any number: ");
        N = sc.nextInt();
        for(x = 1; p < N; x++){
            p = x * (x+1);
            if(p == N){
                System.out.print("\nPronic Number.");
                break;
            }
        }
        if(p > N){
            System.out.print("\nNot a Pronic number.");
        }
    }
}
```

N = 56
p = 2 6 12 20 30 42
x = 1 2 3 4 5 6 7

Pronic Number

N = 28
p = 2 6 12 20 30
x = 1 2 3 4 5

Not a Pronic Number.

Write a program to input a number. Check and display whether it is a Niven number or not. (A number is said to be Niven which is divisible by the sum of its digits).
Example: Sample Input 126
Sum of its digits = 1 + 2 + 6 = 9 and 126 is divisible by 9.

System.out.print() — It writes output, current placement of cursor.
And,
System.out.println() — It writes output, current placement of cursor, and shifts the cursor to new line.
And,
System.out.print("\n") — Shift the cursor to new-line.
But what if we use...
System.out.println("\n"); — Shift the cursor to new-line, again shifts the cursor to new line.
?????

Q23. (Number Problem – Logic based)
Write a Java program to input a number and check whether it is a **special number** or not.
(A special number is a number in which the sum of the factorial of its digits equals the number itself. Example: 145 → 1! + 4! + 5! = 145)

```
import java.util.*;

public class Q23{
    public static void main(String args[]){
        System.out.print("\f");
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter any number: ");
        int n = sc.nextInt();
        int sum=0, f=1, t, d;
        for(t=n; t>0; t/=10){
            d = t%10;
            for( ; d>1; d--){
                f = f*d;
            }
```

```
            f = f*d;
        }
        sum = sum + f;
        f = 1;
    }
    if (sum == n){
        System.out.print("\n Special Number.");
    }
    else {
        System.out.print ("\n Not a Special Number.");
    }
}
}
```

1! = 1
2! = 2 x 1!
3! = 3 x 2!
4! = 4 x 3!
5! = 5 x 4!
6! = 6 x 5!

```
package Looping;


/**
 * Write a description of class Q8 here.
 *
 * @author (your name)
 * @version (a version number or a date)
 */
public class Q8
{
    public static void main(String args[]){
        int i, j, n = 5;
        System.out.print("\f");
        for(i = 1; i <= n; i++){
            for(j = i; j >= 1;j--){
                System.out.print(j);
            }
            System.out.print("\n");
        }
        for(i-=2;i > 0; i--){
            for(j = i; j >0; j--){
                System.out.print(j);
            }
            System.out.print("\n");
        }
    }
}
```

i = 1 2 3 4 5 4 3 2 1 0
j = 1 0 2 1 0 3 2 1 0 4 3 2 1 0 5 4 3 2 1 0
n = 5 j = 4 3 2 1 0 3 2 1 0 2 1 0 1 0 0

output:

```
1
21
321
4321
54321
4321
321
21
1
```

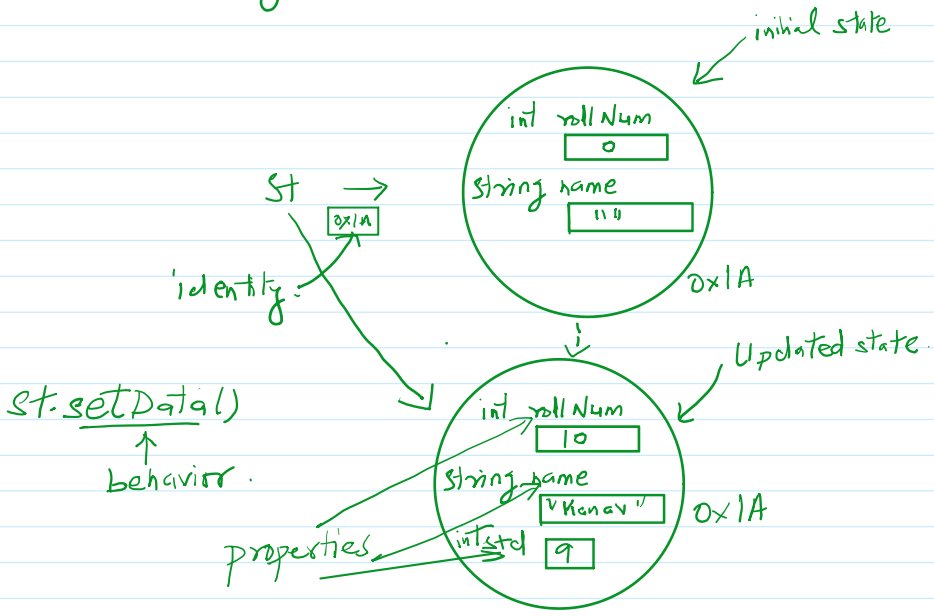
Student st = new Student()

What is an object in Java?

→ It is an instance of a class

What is an object in Java ?

⇒ It is an instance of a class.
⇒ it has its own identity, behavior and state.



Object Relationship with class

"Class is an object factory".

"A class is a blueprint or template that defines the properties and behaviors."

An object is a concrete instance of a class created at runtime.

```
Student st = new Student();
```

Student → class
st → Object
new → Java keyword
Student() → Constructor function. created by the compiler.

it is use to allocate memory at runtime and call Constructor function to initialize the object.

```
public class XMen{
    int id;
    String name;
    public void setData(){
    }
```

id, name → data
setData() → function or method

public void setData() { ← function or method.

Sc

}

public void displayData(){ ← function or method.

...

}

}

Encapsulation binds data (variables) and code (methods) together as a single unit.

Polymorphism

multiple - forms